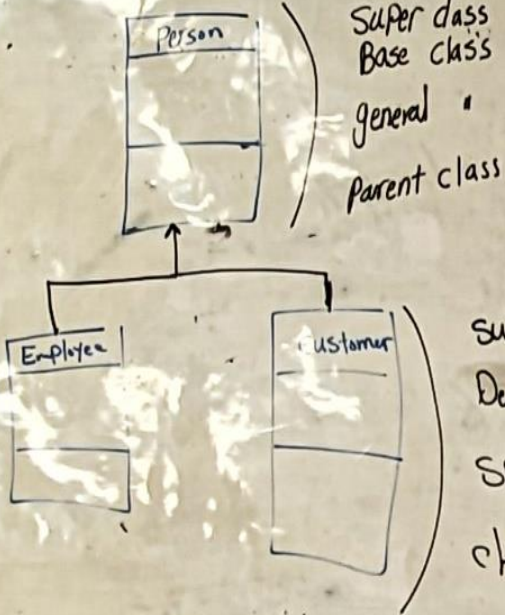


inheritance



Super class
Base class
general
parent class

Relationship $\Rightarrow$ is a

Ex: Employee is a Person
customer is a Person

class

→ Data fields ✓

→ ~~Constructors~~

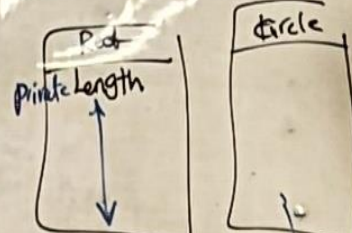
→ Methods ✓

Constructors are not inherited.

Subclasses
Derived classes
Special classes
children classes

Syntax:

Public class Employee extends Person



do len

}

Public class Person

```
{  
    public String name;  
    public int age;  
    public String address;  
    public String nationality;  
}
```

Public Person ()

```
{  
    (String name, int age,  
    String address, String nationality)  
}
```

```
{  
    this.name = name;  
    this.age = age;  
    this.address = address;  
    this.nationality = nationality;  
}
```

Public void setage (int age)

```
{  
    this.age = age;  
}
```

Public int getage ()

```
{  
    return age;  
}
```

Public class Student extends Person

```
{  
    ..  
}
```

Public class test

```
{  
    Public static void main (String[] args)  
}
```

Student S = new student ();

S.setage(21);

S.getage();

5

name: null
age: ~~21~~
address: null
nationality: null

21

Parameter

this.age = age

Current obj

Public class Person

```
public String name;  
public int age;  
public String address;  
public String nationality;
```

```
public Person() {  
    Sy = ("Hello");  
}
```

```
public Person(String name, int age, String address, String nationality) {  
    this.name = name;  
    this.age = age;  
    this.address = address;  
    this.nationality = nationality;  
}
```

```
public void setAge(int age) {  
    this.age = age;  
}  
public int getAge() {  
    return age;  
}
```

Public class Student extends Person

```
private int Studlev;  
private String Specialization;  
private double gpa;  
public Student() {  
    Sy = ("lecture");  
}
```

```
public Student(int studlev, String specialization, double gpa) {  
    this.name = name;  
    this.age = age;  
    this.address = address;  
    this.nationality = nationality;  
    this.Studlev = studlev;  
    this.Specialization = specialization;  
    this.gpa = gpa;  
}
```

```
public void setGpa(double gpa) {  
    this.gpa = gpa;  
}  
public double getGpa() {  
    return gpa;  
}
```

Public class test

```
public static void main (String[] args) {  
    Student S1 = new Student("Ali", 21, "Alex", "Egyptian");  
    S1.setAge(23);  
    S1.setGpa(2.1);  
    Sy = (S1.getGpa());  
    Student S2 = new Student();  
}
```

```
Sy = (S1.getGpa());  
Student S2 = new Student();
```

Memory

S1	Name: Ali
	age: 21-23
	address: Alex
	nationality: Egyptian
	Studlev: 1
	Specialization: AI
	gpa: 2.1

is used to call the constructor of the superclass

2.1 Hello lecture